THE CATEGORY OF QUASI-COHERENT SHEAVES ON AN ALGEBRAIC SPACE IS CLOSED UNDER SUMS, COLIMITS, KERNELS, TENSOR PRODUCTS AND INTERNAL HOMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction through a surjective etale cover by a scheme with the detection criterion for quasi-coherence and the commutation of restriction with limits and colimits, the four additive properties transported through the commutation with restriction, and the treatment of tensor products and internal homs through the commutation with localization.

Editorial note. The label lemma-properties-quasi-coherent names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source spaces-properties.tex –label lemma-properties-quasi-coherent –start 5657 –end 5772 –local-id 28042 returned RESULT=OK (L28042 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-spaces-properties/candidates-halfA.json (fourth pool, pre-assigned by the Lead on 2026-10-13).

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The corresponding statement for schemes. Descent, the lemma listing the same six properties of the category of quasi-coherent sheaves on a scheme. Supplies the six properties in the case where the algebraic space is a scheme.
- Detection of quasi-coherence after restriction to an etale cover. Same chapter, the lemma that a module on an algebraic space is quasi-coherent exactly when its restriction to a scheme etale over it is quasi-coherent. Supplies the criterion by which the six properties are transported from the cover back to the space.
- Restriction along an etale morphism commutes with all limits and colimits. Modules on Sites, the section on localization showing that restriction has both adjoints, hence commutes with limits and colimits; Same chapter, the lemma that the pullback of a quasi-coherent module is quasi-coherent. Supplies the commutation of the restriction functor with the operations appearing in the six properties and the commutativity of the pullback diagram.

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Commutation of the internal hom with localization. Modules on Sites, the lemma that formation of internal hom commutes with restriction. Supplies the identity used in the last of the six properties.

Source reference. Original lemma 0GNB: sources/stacks-chow/tags.txt, source lines 20786–20786. Prerequisite pointer invoked at proof line(s) 5684; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 04M1: sources/stacks-spaces-properties/pinned-original-source.tex, source lines 4733–4733. Prerequisite pointer invoked at proof line(s) 5768; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 03LW: sources/stacks-spaces-properties/pinned-original-source.tex, source lines 4629–4629. Prerequisite pointer invoked at proof line(s) 5704; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03M0: sources/stacks-spaces-properties/pinned-original-source.tex, source lines 5609–5609. Prerequisite pointer invoked at proof line(s) 5712; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 01BG: sources/stacks-spaces-properties/pinned-original-source.tex, source lines 5461–5461. Prerequisite pointer invoked at proof line(s) 5694; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 00XZ: sources/stacks-spaces-properties/pinned-original-source.tex, source lines 4692–4692. Prerequisite pointer invoked at proof line(s) 5767; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 05VR: sources/stacks-spaces-properties/pinned-original-source.tex, source lines 5784–5784. Prerequisite pointer invoked at proof line(s) 5675; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0E8H: sources/stacks-chow/tags.txt, source lines 17817–17817. Prerequisite pointer invoked at proof line(s) 5770; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 03DH: sources/stacks-chow/tags.txt, source lines 4420–4420. Prerequisite pointer invoked at proof line(s) 5707; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. The category $QCoh(\mathcal{O}_X)$ of quasi-coherent sheaves on X has the following properties:*

(1) *Any direct sum of quasi-coherent sheaves is quasi-coherent.*

(2) *Any colimit of quasi-coherent sheaves is quasi-coherent.*
(3) *The kernel and cokernel of a morphism of quasi-coherent sheaves is quasi-coherent.*
(4) *Given a short exact sequence of $\mathcal{O}_X$-modules $0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to 0$ if two out of three are quasi-coherent so is the third.*
(5) *Given two quasi-coherent $\mathcal{O}_X$-modules the tensor product is quasi-coherent.*
(6) *Given two quasi-coherent $\mathcal{O}_X$-modules $\mathcal{F}$, $\mathcal{G}$ such that $\mathcal{F}$ is of finite presentation (see Section 05VR), then the internal hom $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ is quasi-coherent.*

Proof. If X is a scheme, then this is Descent, Lemma 0GNB. We will reduce the lemma to this case by étale localization.

Choose a scheme U and a surjective étale morphism $\varphi : U \to X$. Our notation will be that $Mod(\mathcal{O}_U) = Mod(U_{étale}, \mathcal{O}_U)$ and $QCoh(\mathcal{O}_U) = QCoh(U_{étale}, \mathcal{O}_U)$; in other words, even though U is a scheme we think of quasi-coherent modules on U as modules on the small étale site of U. By Lemma 01BG we have a commutative diagram

$$\begin{array}{ccc} QCoh(\mathcal{O}_X) & \xrightarrow{\varphi^*} & QCoh(\mathcal{O}_U) \\ \downarrow & & \downarrow \\ Mod(\mathcal{O}_X) & \xrightarrow{\varphi^*} & Mod(\mathcal{O}_U) \end{array}$$

The bottom horizontal arrow is the restriction functor (03LW) $\mathcal{G} \mapsto \mathcal{G}|_{U_{étale}}$. This functor has both a left adjoint and a right adjoint, see Modules on Sites, Section 03DH, hence commutes with all limits and colimits. Moreover, we know that an object of $Mod(\mathcal{O}_X)$ is in $QCoh(\mathcal{O}_X)$ if and only if its restriction to U is in $QCoh(\mathcal{O}_U)$, see Lemma 03M0. With these preliminaries out of the way we can start the proof.

Proof of (1). Let $\mathcal{F}_i$, $i \in I$ be a family of quasi-coherent $\mathcal{O}_X$-modules. By the discussion above we have

$$\Big(\bigoplus \mathcal{F}_i\Big)|_{U_{étale}} = \bigoplus \mathcal{F}_i|_{U_{étale}}$$

Each of the modules $\mathcal{F}_i|_{U_{étale}}$ is quasi-coherent. Hence the direct sum is quasi-coherent by the case of schemes. Hence $\bigoplus \mathcal{F}_i$ is quasi-coherent as a module restricting to a quasi-coherent module on U.

Proof of (2). Let $\mathcal{I} \to QCoh(\mathcal{O}_X)$, $i \mapsto \mathcal{F}_i$ be a diagram. Then

$$(\operatorname{colim} \mathcal{F}_i)|_{U_{étale}} = \operatorname{colim} \mathcal{F}_i|_{U_{étale}}$$

by the discussion above and we conclude in the same manner.

Proof of (3). Let $a : \mathcal{F} \to \mathcal{F}'$ be an arrow of $QCoh(\mathcal{O}_X)$. Then we have $\operatorname{Ker}(a)|_{U_{étale}} = \operatorname{Ker}(a|_{U_{étale}})$ and $\operatorname{Coker}(a)|_{U_{étale}} = \operatorname{Coker}(a|_{U_{étale}})$ and we conclude in the same manner.

Proof of (4). The restriction $0 \to \mathcal{F}_1|_{U_{étale}} \to \mathcal{F}_2|_{U_{étale}} \to \mathcal{F}_3|_{U_{étale}} \to 0$ is short exact. Hence we have the 2-out-of-3 property for this sequence and we conclude as before.

Proof of (5). Let $\mathcal{F}$ and $\mathcal{G}$ be in $QCoh(\mathcal{O}_X)$. Then we have

$$(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})_{U_{étale}} = \mathcal{F}|_{U_{étale}} \otimes_{\mathcal{O}_U} \mathcal{G}|_{U_{étale}}$$

and we conclude as before.

Proof of (6). Let $\mathcal{F}$ and $\mathcal{G}$ be in $QCoh(\mathcal{O}_X)$ with $\mathcal{F}$ of finite presentation. We have

$$\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})|_{U_{étale}} = \mathcal{H}om_{\mathcal{O}_U}(\mathcal{F}|_{U_{étale}}, \mathcal{G}|_{U_{étale}})$$

Namely, restriction is a localization, see Section 00XZ, especially formula (04M1)) and formation of internal hom commutes with localization, see Modules on Sites, Lemma 0E8H. Thus we conclude as before. □